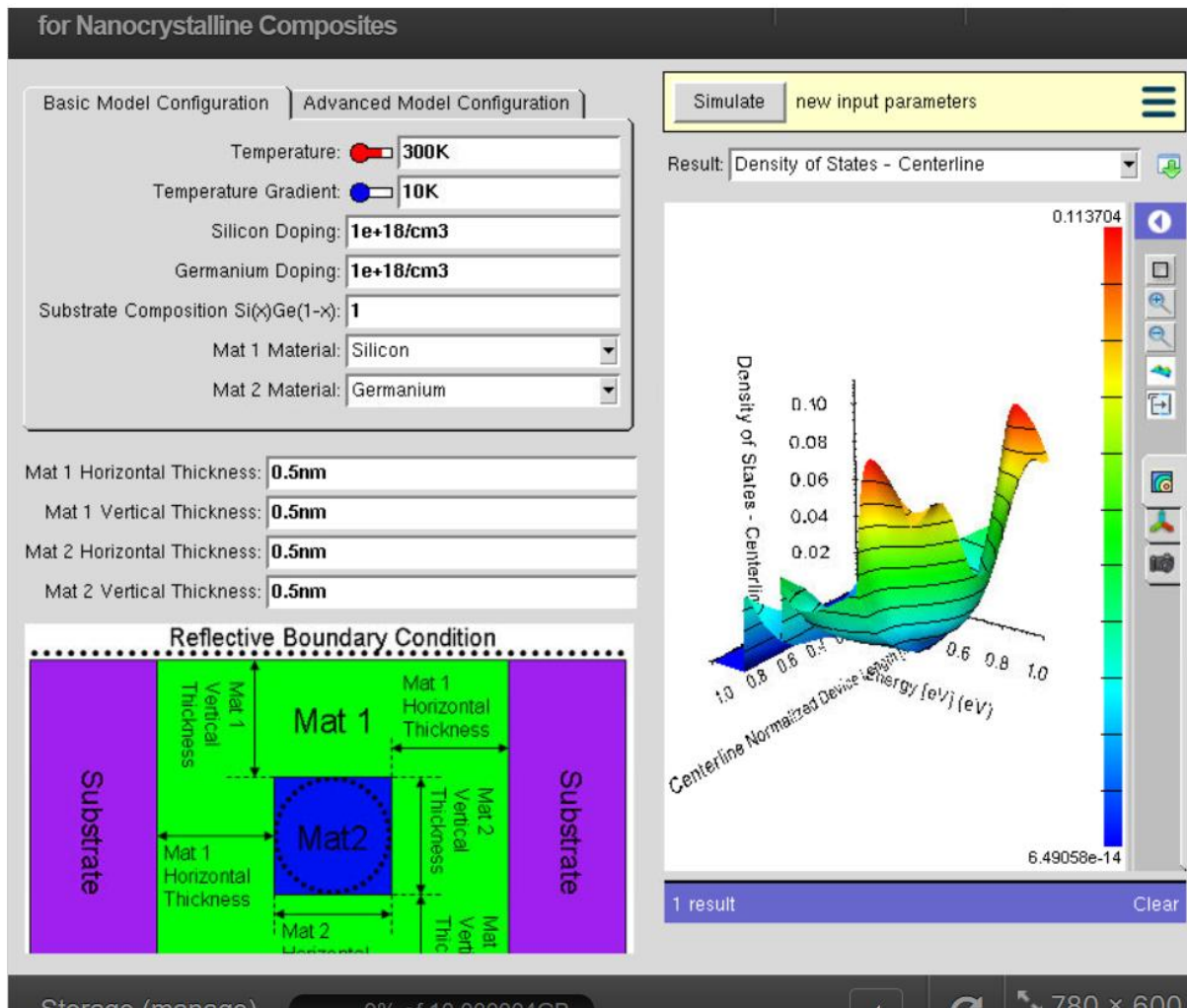


EXP 1: Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness

Case study 1:

Simulate the density of states of quantum well of thickness 0.5 nm

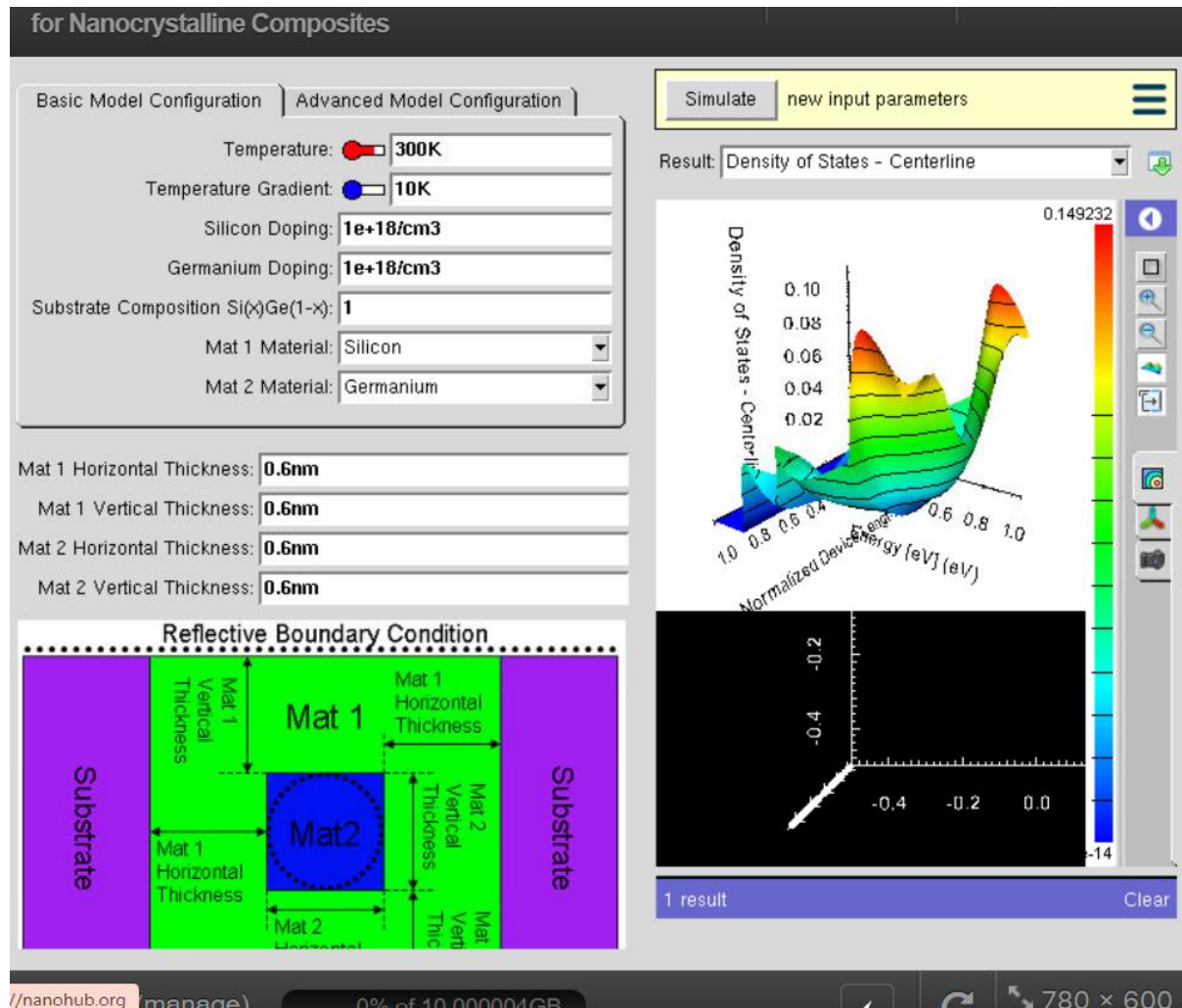


Case Study 1 (0.5 nm thickness):

The graph shows discrete peaks in the density of states due to strong quantum confinement. Thinner wells force carriers into fewer allowed energy levels, resulting in sharper, more widely spaced states

Case Study 2:

Simulate the density of states of quantum well of thickness 0.6 nm



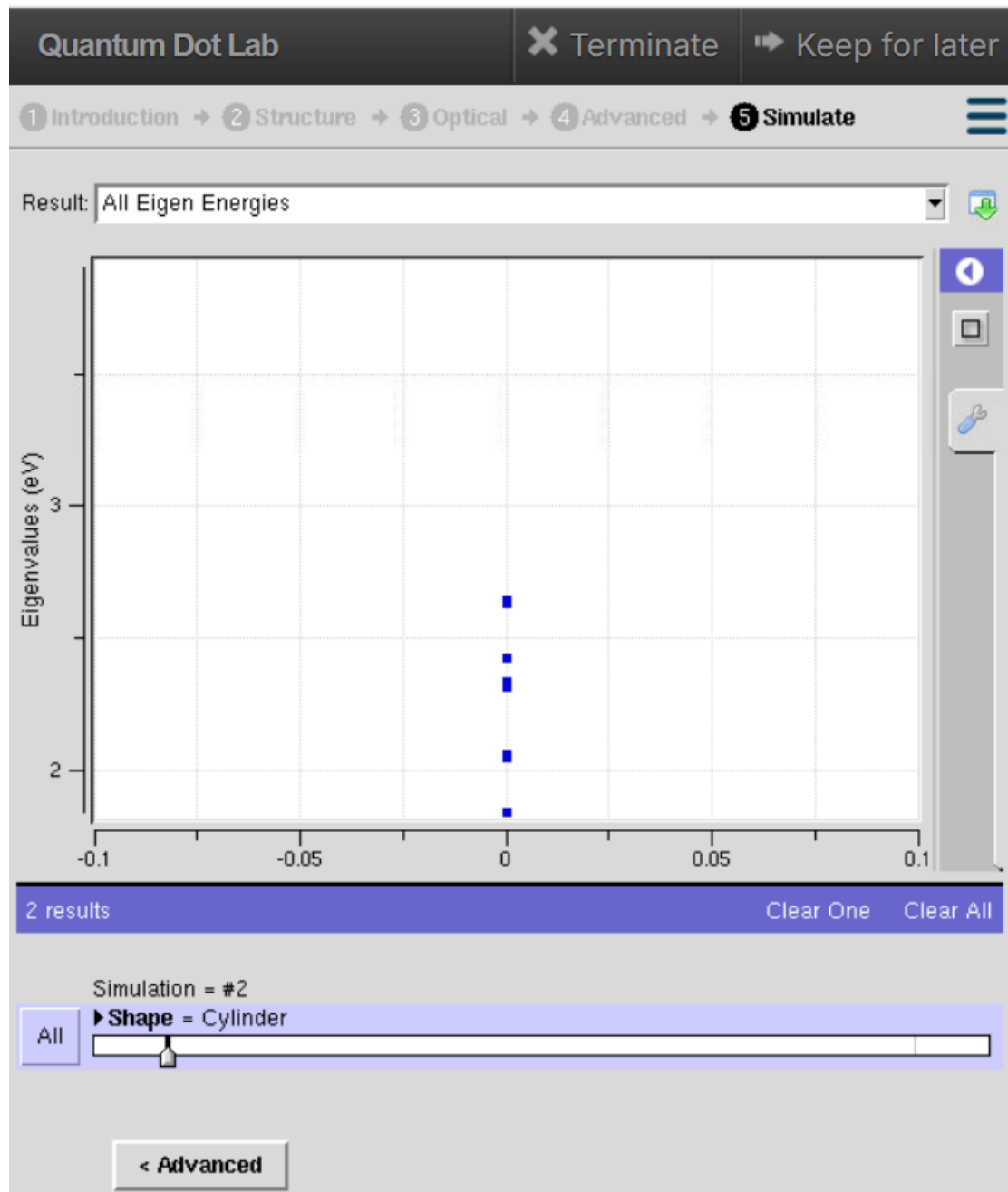
Case Study 2 (0.6 nm thickness):

Here the density of states is slightly smoother compared to 0.5 nm. Increasing thickness reduces confinement, allowing more subbands and lowering the energy spacing between them.

Exp. 2: Simulation of Energy States of Quantum Dot using NanoHub

Case Study 1:

Simulate the energy states of cylindrical quantum dots of $X=15\text{ nm}$, $Y = 10.5\text{ nm}$ and $Z = 8\text{ nm}$.

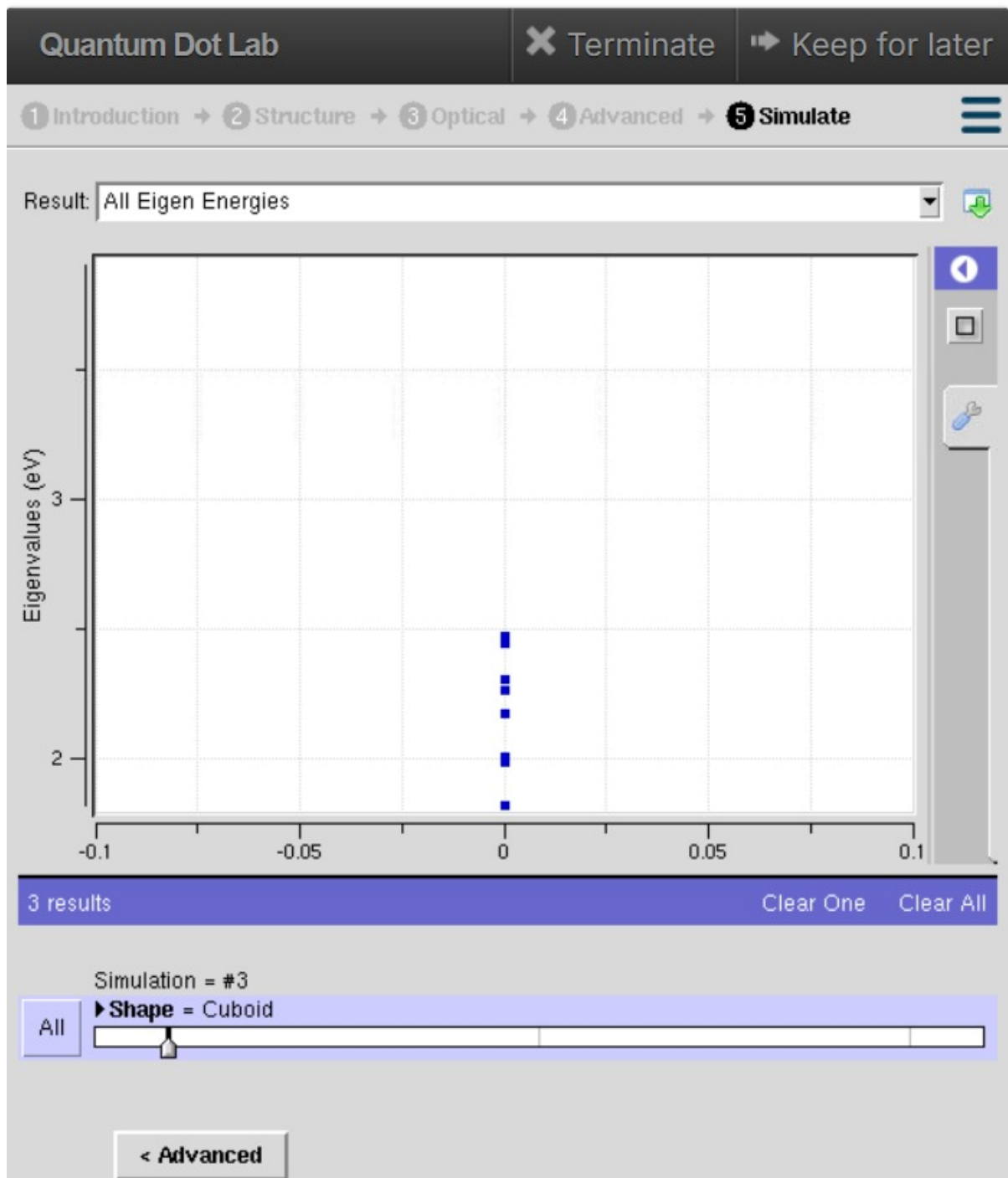


Case Study 1 (Cylindrical dot, $15\text{ nm} \times 10.5\text{ nm} \times 8\text{ nm}$):

The eigenvalue plot shows quantized energy levels. Larger dot dimensions reduce confinement, lowering the energy spacing between states.

Case study 2:

Simulate the energy states of two layer cuboid quantum dots

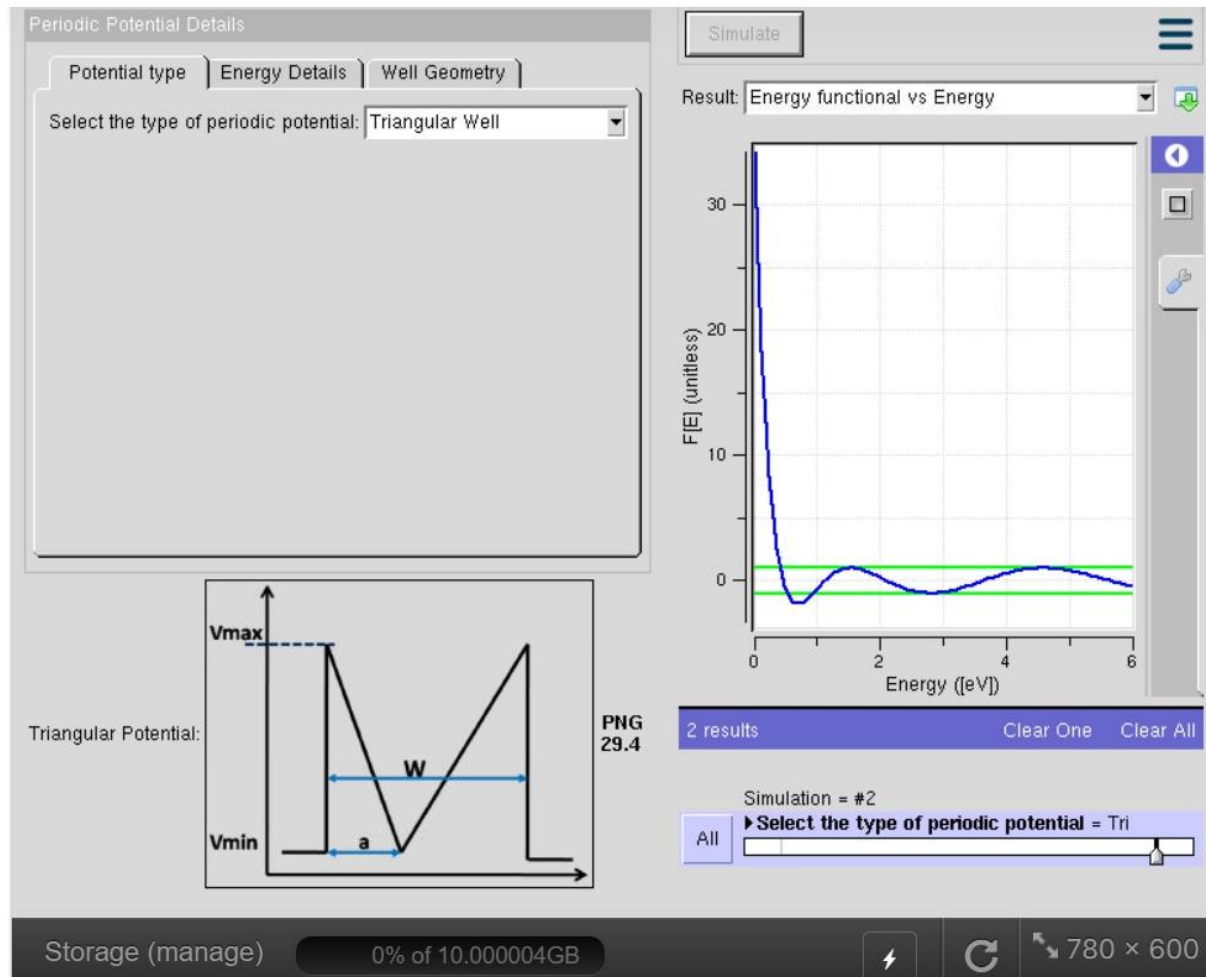


Case Study 2 (Two-layer cuboid dot):

The graph highlights multiple eigenstates due to layered structure. The band offsets between GaAs and InAs create additional confinement effects, shifting energy levels compared to a single dot.

Exp. 3. Kronig Penney Model

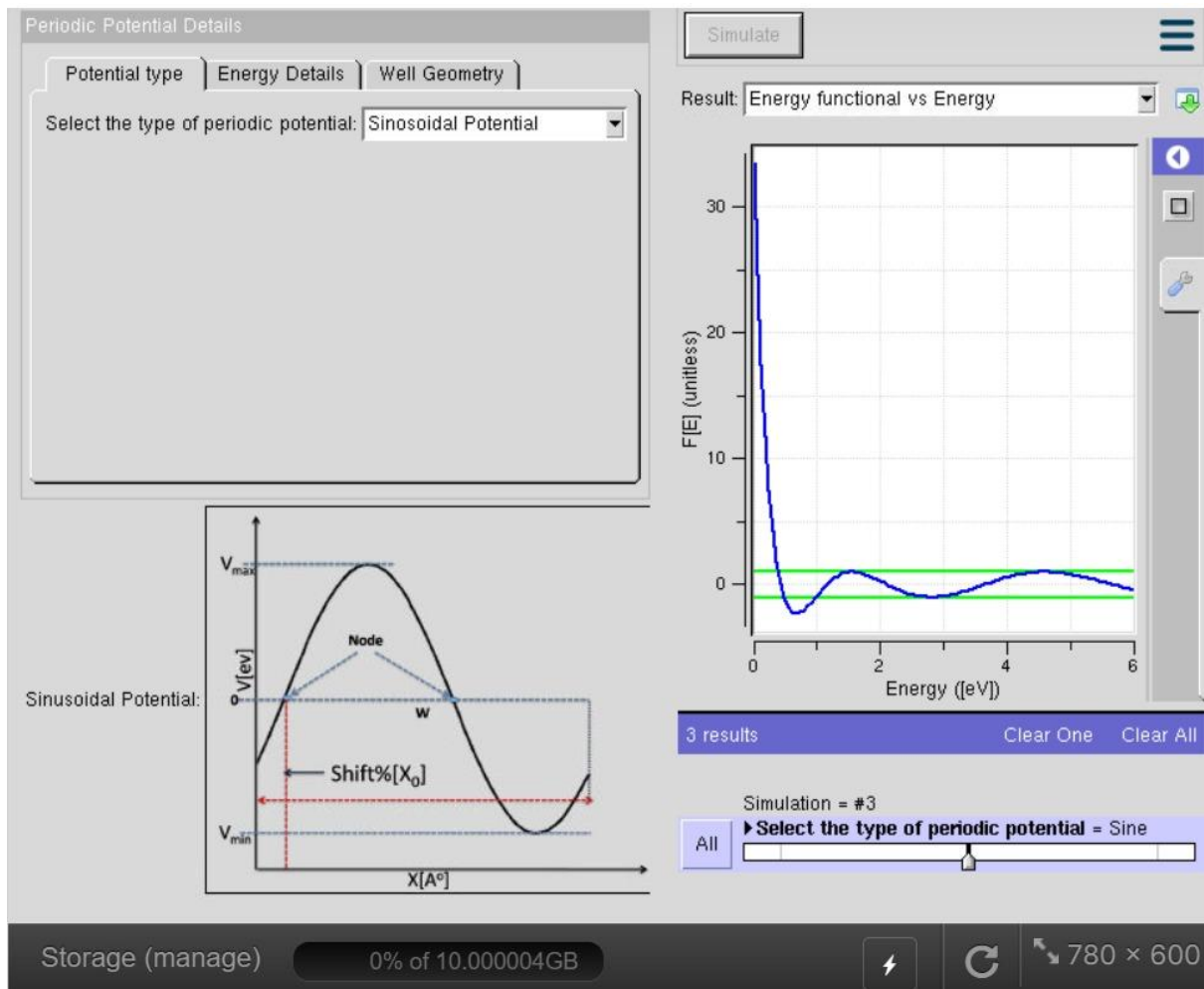
Case Study 1: Simulate the output for Triangular well



Case Study 1 (Triangular well):

The energy functional vs. energy graph shows allowed and forbidden bands. The triangular potential produces narrower bandgaps compared to square wells, reflecting smoother confinement.

Case Study 2: Simulate the output for Sinusoidal Potential

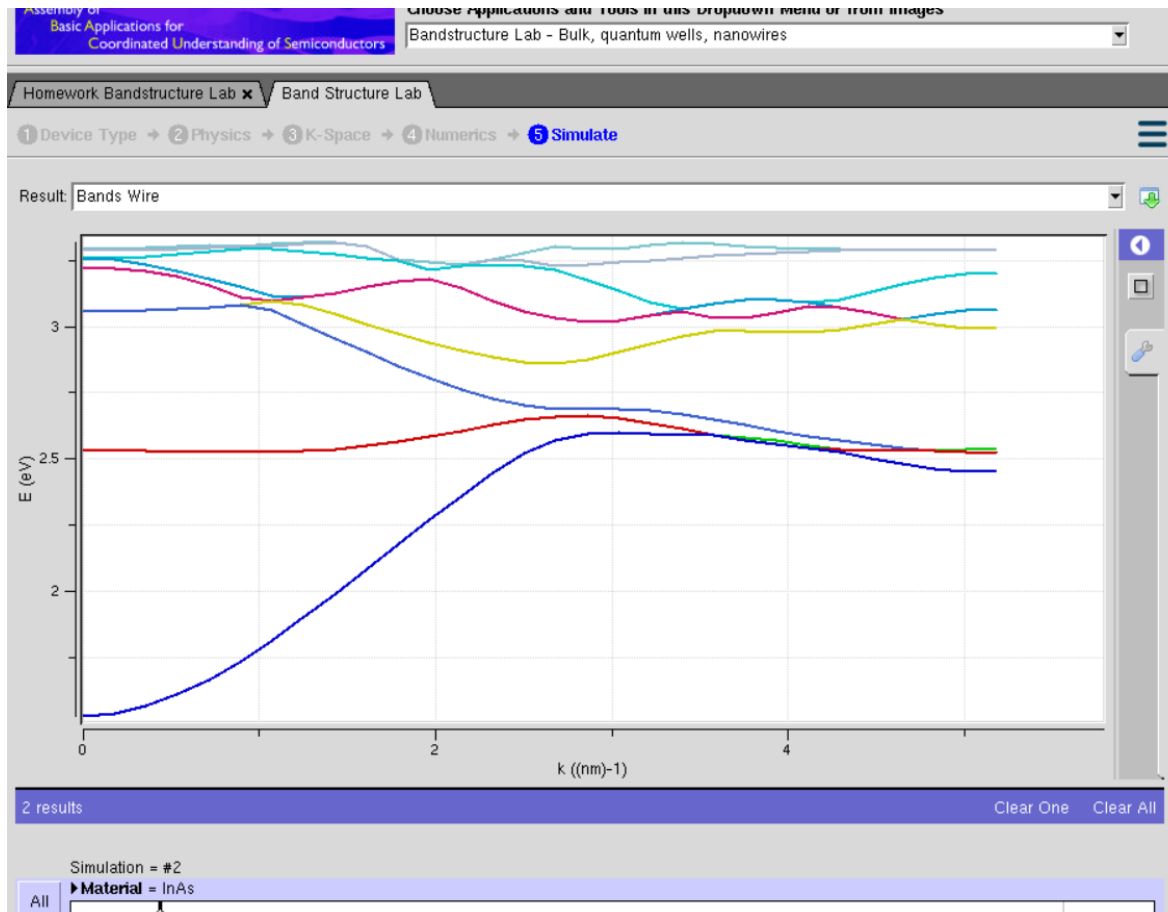


Case Study 2 (Sinusoidal potential):

The sinusoidal potential leads to periodic band structures with smoother transitions. The graph demonstrates how potential shape influences band dispersion and effective electron mass.

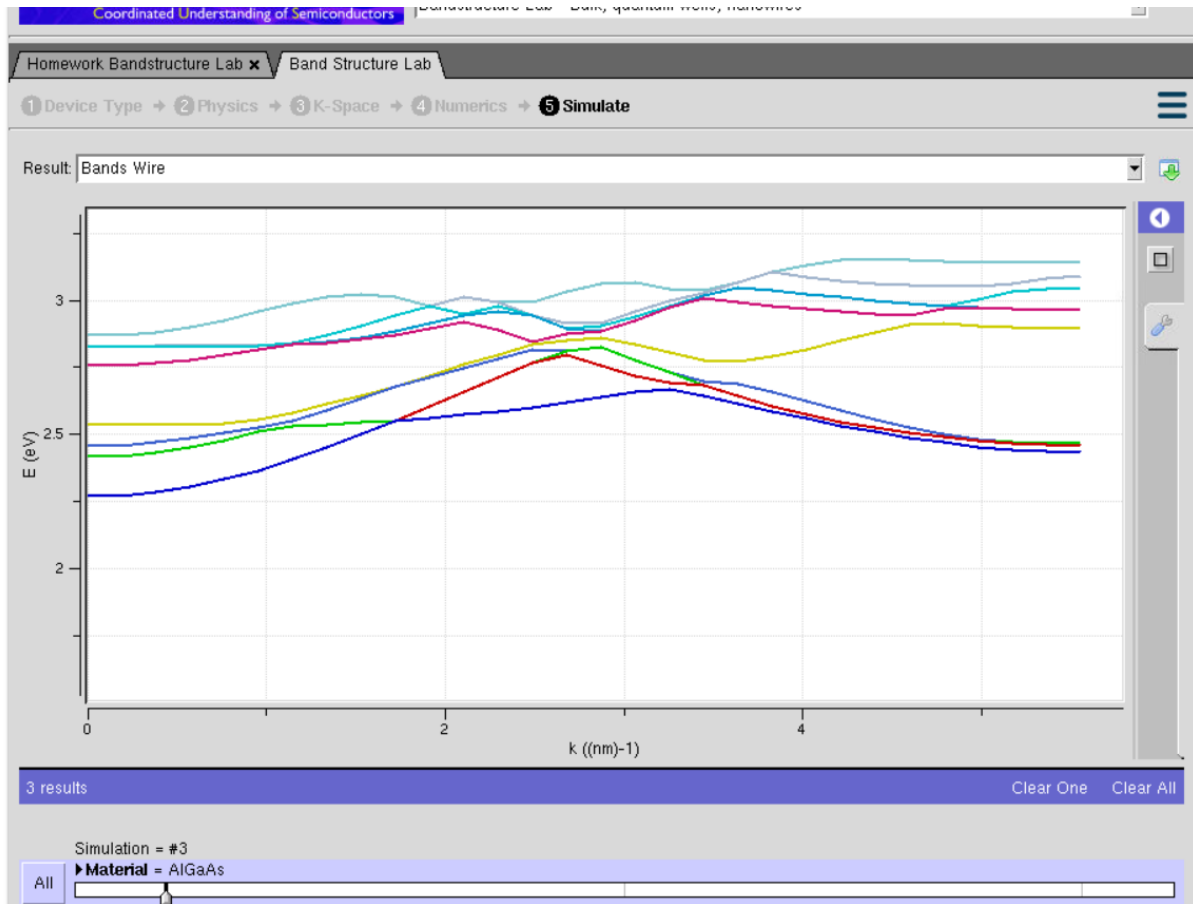
Exp. 4: Band Structure Simulation for Graphene and CNTs using NanoHub

Case Study 1: Simulate the band-structure of InAs circular nanowire



Case Study 1 (Si material, central bands): Here the bandgap is modified by focusing on central bands. The graph highlights how band selection influences electronic properties.

case Study 2: Simulate the band-structure of AlGaAs circular nanowire

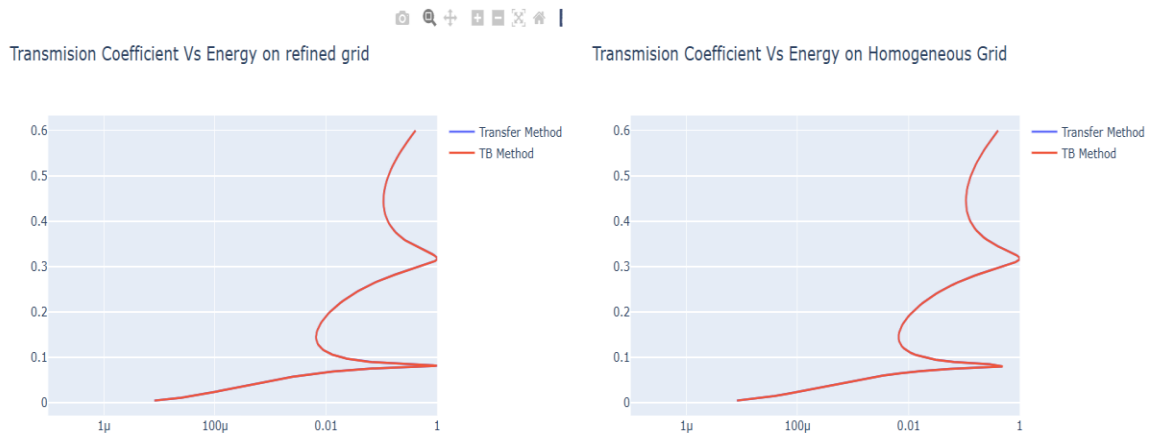


Case Study 2: Germanium's smaller bandgap shifts the dispersion curves. The graph shows how material choice alters conduction and valence band positions.

Exp. 5. Demonstration of Quantum Tunneling

Case study 1 :

Analysis the transmission co-efficient Vs Energy characteristics on refined grid

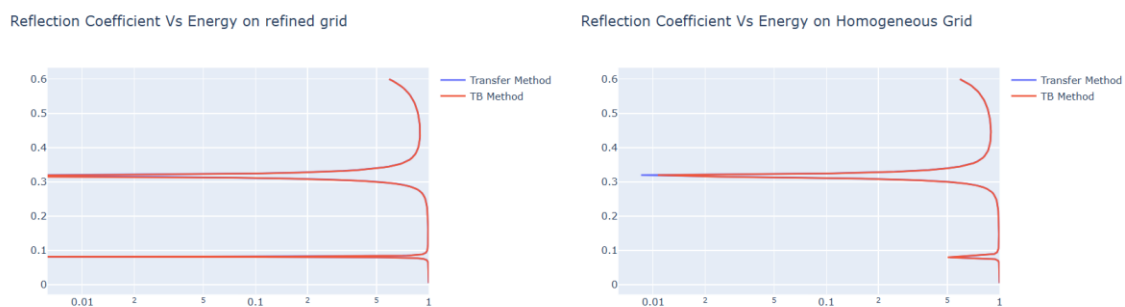


Case Study 1 (Transmission Coefficient vs. Energy on refined vs. homogeneous grid):

The graphs compare transmission probability across energy levels using two numerical methods (Transfer Method and Tight-Binding Method). The refined grid gives smoother, more accurate transmission curves, while the homogeneous grid shows slight deviations. This highlights how grid resolution affects tunneling simulations.

Case study 2 :

Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



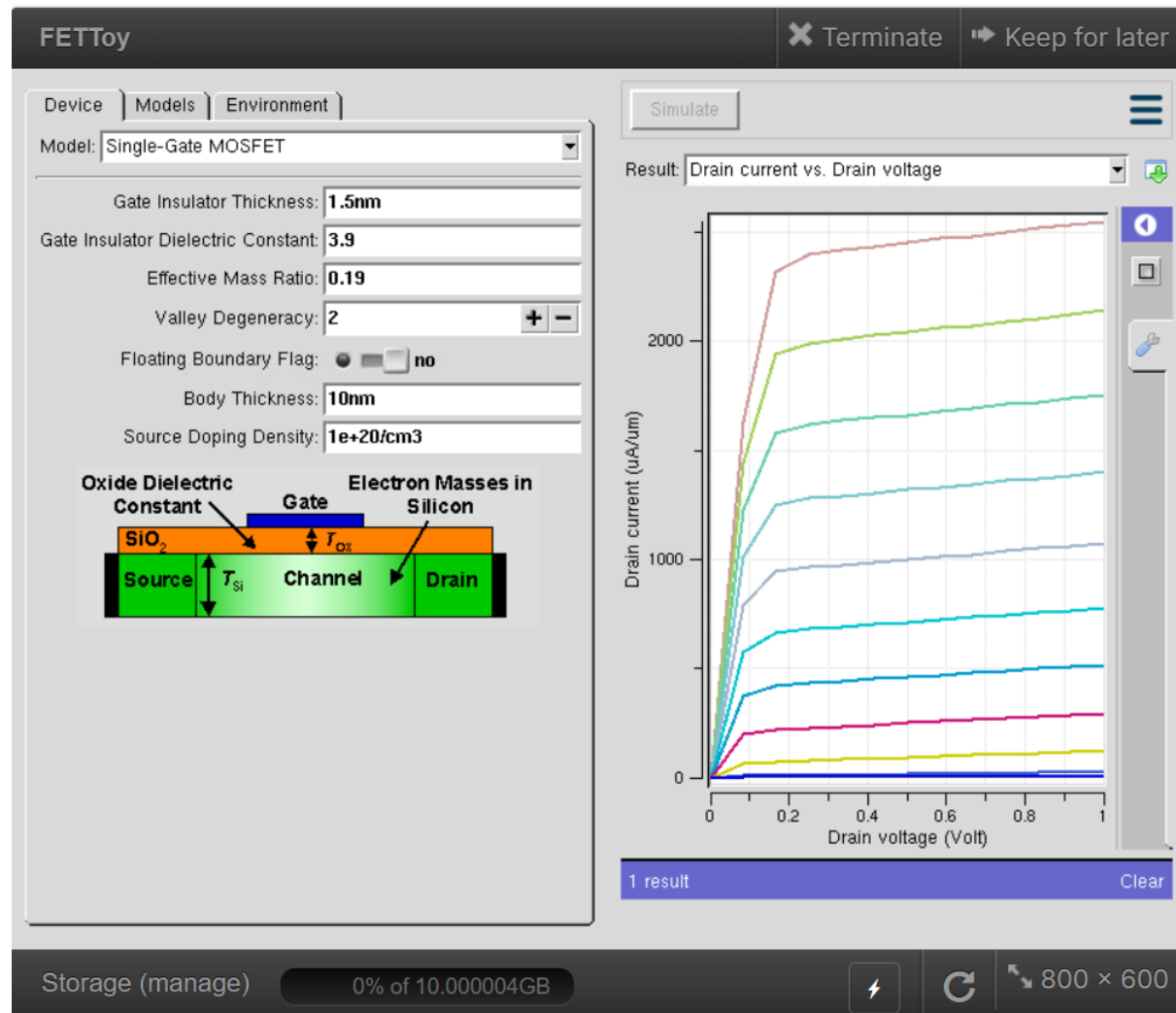
Case Study 2 (Reflection Coefficient vs. Energy on homogeneous grid):

This graph shows how reflection probability varies with particle energy. At lower energies, reflection is high since tunneling probability is low. As energy increases closer

to or above the barrier height, reflection decreases, showing the complementary behavior to transmission.

Exp. 6: MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

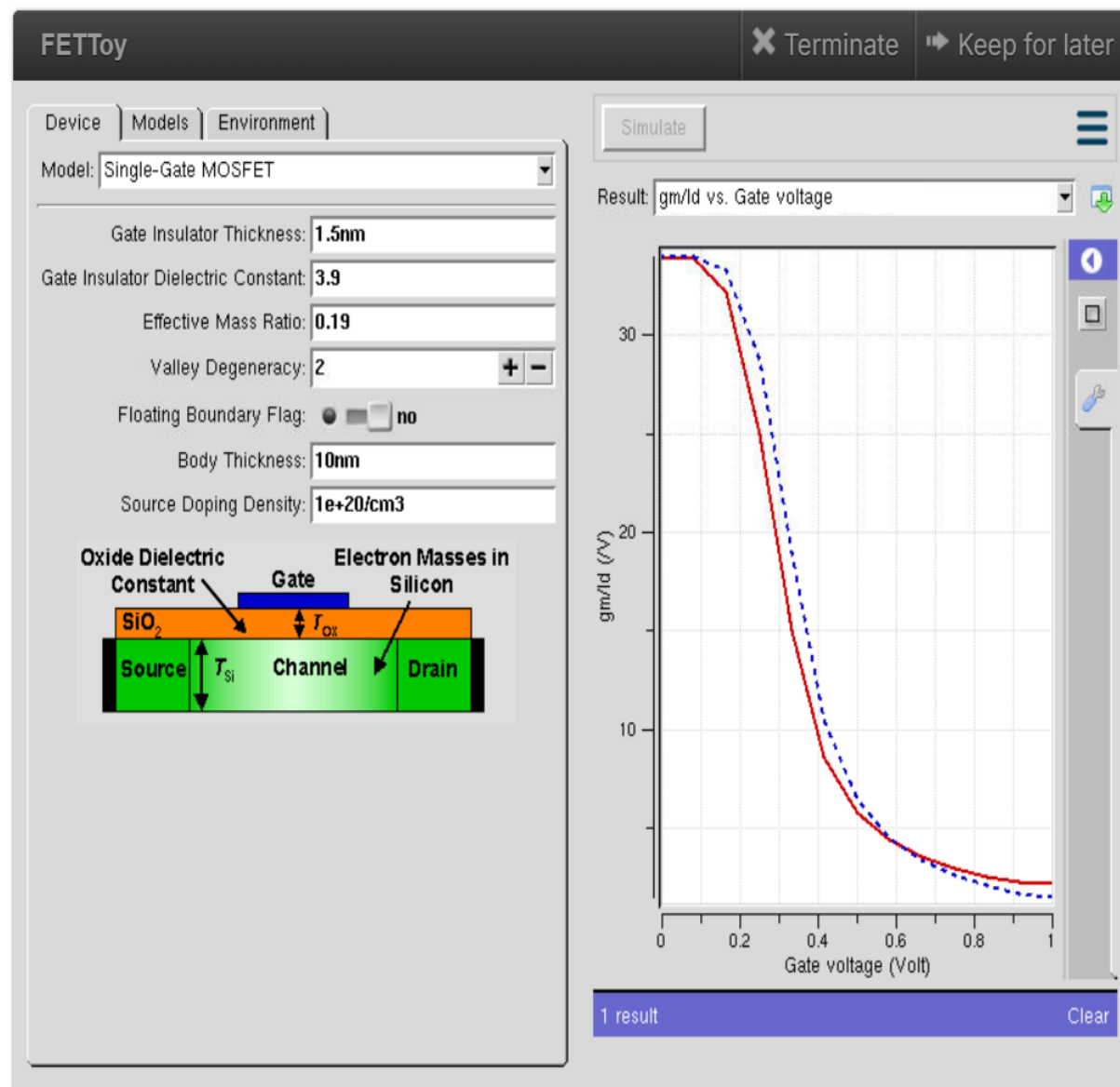
Case study1: Drain current vs Drain Voltage



Case Study 1 (Gate length scaling – I_{ds} vs V_{ds})

Run the I_{ds} – V_{ds} plot for gate lengths of 20 nm and 10 nm. A shorter gate length gives higher drain current due to stronger channel control, but it also increases leakage and short-channel effects. The graph highlights the trade-off between performance and leakage in scaled MOSFETs

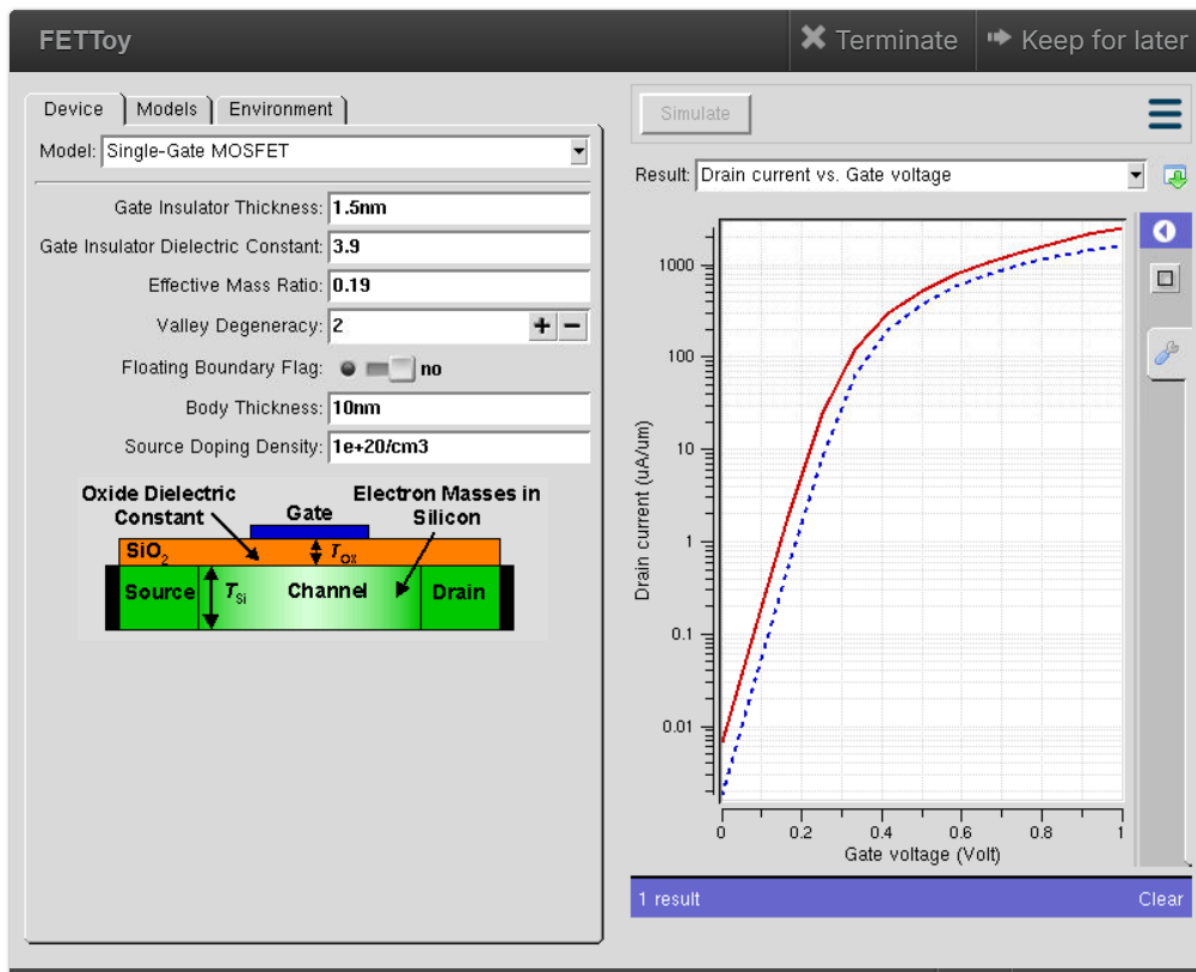
Case study 2: Drain current vs Gate voltage



Case Study 2 (Oxide thickness effect – I_{ds} vs V_{gs})

Run the I_{ds} - V_{gs} plot while varying oxide thickness. Increasing oxide thickness reduces gate control, which raises the threshold voltage and lowers current drive. The graph shows how gate dielectric scaling directly impacts switching efficiency.

Case study 3: Drain current vs Gate voltage



Case Study 3 (Subthreshold slope – log I_{ds} vs V_{gs})

Run the semi-log $I_{\text{ds}}-V_{\text{gs}}$ plot for different channel materials. Materials with better electrostatics (like Ge compared to Si) show steeper slopes, meaning improved switching and lower leakage. The graph demonstrates how material choice affects subthreshold swing.